

SRS-XXXX

SOFTWARE REQUIREMENTS SPECIFICATION

Sentinel: A Geopolitical Travel Safety Advisor Website

for:

Travelers Worldwide

Prepared by:



S1 Informatics Study Program

School of Computing

Telkom University

2025

 S1 Informatics Study Program - School of Computing	SRS - Document Number		60
	Revision	C	<i>Date: November 19, 2025</i>

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1. Revision History

Revision	Description
A	This revision expands the system's functional scope. The Functional Requirements (FRs) have been increased from 7 to 16. A new User Class, "Affiliated Journalist," has been added. The Use Case Diagram (UCD) and associated Use Case Scenarios were revised to correct deficiencies and reflect the updated requirements.
B	This revision focused on enhancing UML model accuracy and completing document sections. Key changes include updating the Use Case Diagram to ensure actors and use cases match the defined User Classes and Functional Requirements (FRs), adding FR codes for traceability, and refining the Class Diagram to correct an improper user generalization. New Activity Diagrams were also added. Content was completed for Chapter 4.1 (User Interface) with initial screen images, Appendix A was populated, and IEEE standard references were added to section 1.4. Finally, a team contribution list was included as a document attachment.
C	This revision updates formatting, diagram accuracy, and requirement specification details. First, formatting inconsistencies regarding page numbers and section numbering must be corrected, and the Document Scope (Section 1.2) needs refinement. Diagrammatic changes are required, specifically correcting attributes in the Class Diagram and fixing errors in Activity Diagrams. Regarding Functional Requirements, FR-05 is identified as non-functional and must be removed along with its associated use case scenario and activity diagram; additionally, all remaining functional requirements must be categorized by type (Must-have, Satisfier, Delighter). Actor specification in Use Case Scenario #13 requires specific text edits. Finally, Section 4.1 is added with a reference website screenshot.

INDEX	-	A	B	C
DATE		October 20, 2025	November 10, 2025	November 17, 2025
Written by		Respective members	Respective members	Respective members
Reviewed by		Olaza [REDACTED] [REDACTED]	Olaza [REDACTED] [REDACTED]	Olaza [REDACTED] [REDACTED]
Approved by		Olaza [REDACTED] [REDACTED]	Olaza [REDACTED] [REDACTED]	Olaza [REDACTED] [REDACTED]

2. List of Changed Pages

Page	Revision	Page	Revision
5	1.2 Document Scope	55	Updated Class Diagram
10-15	Moving of previously FR-05 (UI-related) to Non Functional	56	Figure added to 4.1 User Interface
10-13	Categorized FRs (Must Have, Satisfier, and Delighter)	56-57	Updated FRs information on Section 4
16	Updated Use Case Diagram	58	Updated FRs information on Appendix A
37-39	Added actor specification on Use Case Scenario #13 flows.	60	Updated FRs information on Attachment (Contribution)
45-54	Updated Activity Diagrams		

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1. Introduction

1.1 Document Purpose

This Software Requirements Specification (SRS) document defines the software requirements for Sentinel: A Geopolitical Travel Safety Advisor Website, currently at its initial release. This document specifies user interactions and constraints along with the functionalities. It provides reference for testers, developers, and stakeholders to ensure that the intended objectives are met with the delivered software.

The major or primary purpose of the SRS is to describe the features that enable government entities, travel agencies, travelers, and affiliated journalists to access verified and updated geopolitical and travel safety information. This SRS also covers the whole Sentinel platform, which includes user account management, account management, travel advisories, safety alerts, and content verification workflows.

1.2 Document Scope

Sentinel is a web-based platform designed to provide travelers with reliable, up-to-date information regarding safety and geopolitical risks worldwide. The system aggregates verified public data from government travel advisories, international news sources, and public health organizations. Its primary objective is to enhance travel awareness, highlight potential risks, and support users in making safe and informed decisions. Sentinel functions strictly as an informational platform, serving both as a safety advisor and a decision-support tool.

Conventions Used in This SRS:

- **Font and Highlighting:** Standards are applied consistently across requirement descriptions for clarity and readability.
- **Requirement Classification:** Requirements are categorized as “Must Have,” “Satisfier,” and “Delighter,” indicating priority and expected user impact.
- **Priority Inheritance:** Unless explicitly stated otherwise, higher-level requirements pass their priority level down to detailed functional requirements.
- **Terminology:** Terms such as “traveler,” “travel agency partner,” “administrator,” “risk expert,” and “affiliated journalist” are used consistently to represent distinct user groups within the system.

This SRS serves as the foundational document for the development, testing, and maintenance of the Sentinel platform. It ensures that all functional and non-functional requirements are clearly defined, traceable, and achievable.

1.3 Definitions, Acronyms, and Abbreviations

This document defines several key terms and acronyms essential to understanding the requirements for the Sentinel platform. Sentinel is the name of the global travel safety advisor website itself, designed to provide essential safety and security intelligence to users. The document you are reading is the SRS (Software Requirements Specification), and the platform's individual features are described as FRs (Functional Requirements). The system serves several main user classes: the Traveler (the primary user seeking destination safety information), the Travel Agency Partner (using insights to advise clients), the Administrator (managing system integrity and accounts), and the Risk Expert (validating and classifying geopolitical data). The system must also provide an intuitive UI (User Interface) to ensure easy navigation for all users.

1.4 References

- [1] T. S. Gov, “Travel Advisories | Travel.State.gov,” *State.gov*, 2025. Available: <https://travel.state.gov/en/international-travel/travel-advisories.html>
- [2] World Travel Guide, “World Travel Guide,” *World Travel Guide*, 2019. Available: <https://www.worldtravelguide.net/>
- [3] CnTraveler, “Condé Nast Traveler - Travel Reviews, News, Guides & Tips,” *Condé Nast Traveler*, 2019. Available: <https://www.cntraveler.com/>

2. General Software Description

2.1 Software Statement of Objective

Our application with the name of “Sentinel” is an intentional and new, self contained product designed to address and solve the problem of travelers being unaware of localized dangers and geopolitical realities and dangers as well of their destination. The general purpose of sentinel is to serve as a centralized resource that provides the best essential safety and security intelligence to help users make safer decisions when travelling. The scope of this software is strictly to raise awareness by presenting trusted, easy to understand, and publicly available data from vetted sources, such as official government travel advisories, verified international news, and public health organization alerts. The platform’s focus is informational so it shall not provide travel bookings, logistical plannings, etc. And the main business objectives are to help travelers become more aware, well informed, and better prepared before their journey while also reducing travel related risks. And at the end of the day the software’s benefit is to help travelers feel more confident, stay safe, and better prepared through their journeys around the world.

2.2 Software Perspective and Goals

High level business goal :

- 1) Goal 1 : Deliver Vetted, real time safety information
 - Correlated Functional Requirements (FR Code) :
FR-01, FR-02, FR-08, FR-11, FR-12, FR-14
- 2) Goal 2 : Provide an intuitive and supportive user experience
 - Correlated Functional Requirements (FR Code) :
FR-05, FR-06, FR-07
- 3) Goal 3 : Ensure Data Accuracy and System Integrity
 - Correlated Functional Requirements (FR Code) :
FR-10, FR-12, FR-14, FR-16
- 4) Goal 4 : Reduce Traveler-Related Risk
 - Correlated Functional Requirements (FR Code) :
FR-03, FR-04, FR-13, FR-15

2.3 User Classes and Profiles

2.3.1. Traveler

This user represents individuals or small groups who travel domestically or internationally and seek to understand the safety conditions of their destinations. They use “Sentinel” to view real-time geopolitical risk maps, receive alerts, and explore background information on regional issues.

2.3.2. Travel Agency Partner

Travel agencies and consultants use “Sentinel” to access real-time safety insights before recommending destinations to clients. They rely on the platform to assess regional risks, plan secure travel routes, and provide safety briefings. This helps them deliver trusted, data-driven travel advice and enhance client confidence in their services.

2.3.3. Administrator (System Admin)

The administrator is responsible for maintaining the system’s integrity by managing user data, updating the geopolitical database, and ensuring that the information presented on the platform remains accurate and current.

2.3.4. Data Analyst/Risk Expert

This role involves collecting and validating data from official sources such as embassies, global news outlets, and international safety indexes. These users classify and update regional risk levels on the platform.

2.3.5. Affiliated Journalist

This role will do field research in the countries to which they are assigned (preferably where the journalists are local to the country). They will investigate news, interview sources, and present accurate and fair accounts of events to avoid bias, then submit the information for verification.

2.4 Operating Environment

The Sentinel platform operates as a web-based travel safety advisory system, similar to global travel-information services such as the U.S. State Department Travel Advisories, World Travel Guide, and Condé Nast Traveler. In a client environment, it will be accessible via modern web browsers, fully responsive for desktops, tablets, and mobile devices, and will require JavaScript, HTML5, CSS3, and an active internet connection to load dynamic data such as travel advisories and alerts. In a server environment, it will be hosted on a cloud-based server to support high availability and global access. The backend is implemented using a web framework, and Real-time data retrieval from external public sources through approved APIs or scraping pipelines, depending on availability and policy compliance. External interfaces may integrate with public advisories such as government travel warnings, destination safety indexes, cultural guidance, and an interactive world map delivered through a mapping API (Mapbox or Leaflet).

2.5 Software / System Constraints

Inspired by the processes and limitations of real travel-advisory systems like the U.S. State Department website, the platform must comply with the following constraints. First is Data & Content Constraints, where Sentinel can only display publicly accessible data. No confidential or

classified intelligence is allowed; travel information accuracy depends on third-party update frequency, and some regions may lack consistent or real-time data availability. Second, Technical Constraints where pages containing advisory information and interactive maps must load within acceptable performance standards, dependent on the uptime of external travel-advisory platforms for inbound data feeds, and the mapping system requires stable access to mapping services. Third is security & compliance constraints, where it must use HTTPS for encrypted communication, must comply with global data-protection standards (e.g., GDPR) for user information, and cannot modify official advisory data—only aggregate, summarize, and present it. Lastly, the design and platform constraints where it follows responsive web-design practices to support multiple screen sizes and use accessible color schemes to ensure safety indicators.

2.6 Assumptions and Dependencies

Based on how global travel-advisory sites operate, Sentinel assumes the following:

Assumptions

External sources remain reliable: Sentinel assumes that official sources (e.g., travel.state.gov) continue publishing updated travel advisories.

Users have an internet connection: Travel-advisory systems require live data; offline mode will not provide current risk alerts.

Third-party data is accurate and updated regularly: Sentinel verifies but does not independently investigate every global event.

Users understand basic travel terminology: Concepts like “Level 3 Advisory,” “Entry Restrictions,” or “High-Risk Area” are assumed to be understood or can be explained through the UI.

The interactive risk map remains functional: Depends on mapping services and correct geolocation data.

Dependencies

Government advisories for safety alerts.

International travel guides for cultural info and travel background.

Journalism platforms for trend analysis and recent global incidents.

Mapping API providers for displaying geographic risk zones.

Cloud hosting services for availability, server resources, and uptime.

If any of these sources or services become unavailable, certain Sentinel features—such as risk mapping, alert updates, or cultural advisory summaries—may be temporarily limited.

3. Detailed Software Description

3.1 Requirements Descriptions

3.1.1 Functional Requirements

The following table 3.1.1.1 lists the functional requirements.

No.	Requirement Code	Requirement Name	Description	FR Type
1	FR-01	View Destination Information	The system shall display verified and up-to-date safety information upon the user's selection of a destination.	Must Have
2	FR-02	View Safety Alerts and Emergency Contacts	The system shall display recent safety alerts and emergency contacts for the selected destination.	Satisfy
3	FR-03	Display Highlighted Key Safety Points	The system shall highlight key safety points derived from verified data.	Satisfy

4	FR-04	Display Short Safety Tips	The system shall display travel advisories, cultural tips, and awareness notes for the selected destination.	Satisfy
5	FR-05	Save Important Safety Reminders	The system shall allow travelers to save selected safety reminders.	Delighter
6	FR-06	Display Short Safety Quotes	The system shall display awareness-promoting safety quotes on the homepage.	Delighter
7	FR-07	View Interactive Risk Map	The system shall provide an interactive geopolitical risk map with color-coded risk levels.	Must Have
8	FR-08	Register New Account	The system shall allow users to create new accounts as either Traveler or Travel Agency Partner.	Must have

9	FR-09	Manage User Accounts	The system shall allow Administrators to create, update, and delete user accounts.	Must Have
10	FR-10	Search Bar for Destinations	The system shall provide a search bar to find destination information easily.	Must have
11	FR-11	Verifying Information	The system shall enable journalists to submit and experts to verify geopolitical data accuracy.	Must have
12	FR-12	Data Filtering and Sorting	The system shall allow users to filter or sort destinations and news articles by risk level.	Satisfy
13	FR-13	Administrator Data Management	The system shall allow administrators to add, update, or delete database content.	Must have

14	FR-14	Travel Route Assistance	The system shall suggest optimal travel routes based on geopolitical and safety data.	Delighter
15	FR-15	Application Login	The system shall allow registered users to log in using valid credentials.	Must have

Table 3.1.1.1 Functional requirements

3.1.2 Non-Functional Requirements

The following table 3.1.2.1 lists the non-functional requirements.

No.	Quality Criteria	Requirement Code	Description	Category
1	Usability	NFR-01	The system interface must be user-friendly, intuitive, and responsive for various user roles.	User Experience

2	Performance	NFR-02	The system must load destination and safety data within 3 seconds under normal conditions.	Speed
3	Reliability	NFR-03	The system must maintain 99% uptime and handle unexpected input or connection loss gracefully.	Robustness
4	Security	NFR-04	User credentials, reminders, and safety data must be stored securely with encryption.	Protection
5	Maintainability	NFR-05	The codebase must be modular and easily maintainable for adding new destinations or risk data.	Development
6	Scalability	NFR-06	The platform must support increasing user traffic and additional regions without performance loss.	Growth

7	Availability	NFR-07	The system must be accessible 24/7 except during scheduled maintenance.	Uptime
8	Compatibility	NFR-08	The web application must function across major browsers and mobile devices.	Interoperability
9	Accessibility	NFR-09	The interface must be accessible to users with visual impairments (WCAG compliance).	Inclusivity
10	Data Integrity	NFR-10	All safety data displayed must be verified and protected from unauthorized modification.	Data Quality

Table 3.1.2.1 Non-functional requirements

3.2 Analysis Modeling

3.2.1 Use Case Diagram

The following is the Use Case Diagram (UCD).

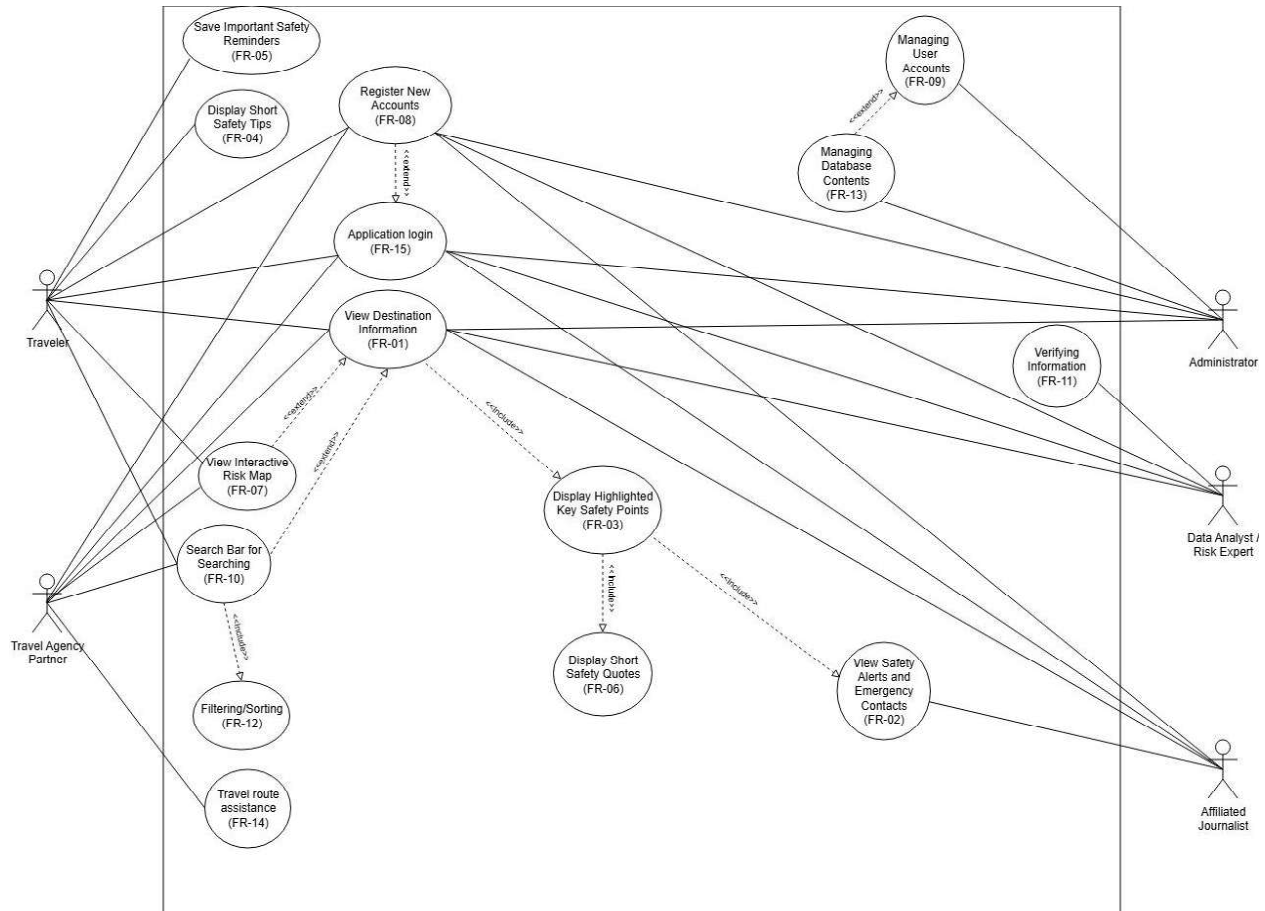


Figure 3.2.1.1 Use Case Diagram

3.2.2 Use Case Scenario

3.2.2.1 Use Case Scenario #1 <View Destination Information>

Use Case Name	View Destination Information
Description	The system displays verified and up-to-date safety information upon the user's selection of a destination, including the calculated risk level and key geopolitical status.

Pre-Condition	The Actor (Traveler or Partner) is logged into the Sentinel system (FR-16). Verified Destination data exists in the database.	
Post-Condition	The Destination's overall safety profile is successfully rendered on the screen (FR-05)	
Main Scenario	Actor	System
	1. The actor uses the search bar (FR-11) or interacts with the map (FR-08) to select a destination.	
		2. System queries the database using the geopolitical identifier (e.g., country code).
		3. System retrieves the core Destination entity and associated status data.
		4. System processes associated <i>RiskReports</i> to calculate the current geopolitical risk level.
	5. The actor views the main profile page.	
		6. Offers an option to view recent alerts and cultural advisories.
Exceptional Scenario (Alternative Flow)	Actor	System
	1. The actor enters a search query.	
		2. Search query yields zero matching destinations.
	3. The actor receives notification.	

		4. The system presents a "No results found" message and suggests related broader regions or similar countries.
	5. The actor refines search.	
		6. System retrieves the new Destination entity (Loop to Step 3 Normal Flow).

3.2.2.2 Usecase Scenario #2 <View Safety Alerts and Emergency Contacts>

Use Case Name	View Safety Alerts and Emergency Contacts	
Description	When a traveler or partner selects a destination, the system displays recent safety alerts from trusted sources and emergency contact details relevant to that location.	
Pre-Condition	The traveler is logged in and has selected a destination.	
Post-Condition	Recent safety alerts and verified emergency contact details are successfully shown.	
Main Scenario	Actor	System
	1. Selects a destination.	
		2. Retrieves recent safety alerts and emergency contacts from verified sources.
	3. Review the displayed alert	
		4. Displays updated safety alerts and trusted emergency contact numbers clearly.
Exceptional Scenario (Alternative flow)	Actor	System
	1. Selects a destination.	

		2. If no recent alerts or contacts are available, displays a message such as “No current alerts” or provides general emergency information.
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3.2.2.3 Usecase Scenario #3 <Display Highlighted Key Safety Points>

Use Case Name	Display Highlighted Key Safety Points	
Description	When an actor views destination details, the system highlights key safety points—summarized risks and actionable advice—derived from verified intelligence (FR-12 results).	
Pre-Condition	The Actor is viewing the Destination Information page (Post-Condition of FR-01). Key safety summaries have been successfully processed and associated with the Destination entity.	
Post-Condition	A dedicated section of the destination profile page displays clear, concise safety points. Normal Flow of event	
Main Scenario	Actor	System
	1. The actor scrolls down the Destination Information page.	
		2. System identifies the required summarized content linked to the destination (FR-03).
		3. System retrieves and formats the key safety points (e.g., "Avoid Night Travel in Region X," "Common Scam: Taxi Overcharging")

	4. The actor reads the highlighted points.	
		5. The system displays the critical safety points in a visually prominent, distinct section (FR-03).
	6. Actor saves one of the points as a reminder (FR-06).	
		7. The system processes the reminder save request.
Exceptional Scenario (Alternative flow)	Actor	System
	1. System attempts to identify summarized content.	
		2. No specific, manually highlighted Key Safety Points exist for the destination yet (content team has not yet generated them).
	3. The actor views the page.	
		4. The system displays a generic "General Safety Information Available Above" message in place of the highlighted section.

3.2.2.4 Usecase Scenario #4 <Display Short Safety Tips>

Use Case Name	Display Short Safety Tips
Description	When a traveler selects a destination, the system shall display travel advisories, including cultural tips, and quick awareness notes.

Pre-Condition	The traveler has selected a destination.	
Post-Condition	Safety tips and awareness notes are displayed on the interface.	
Main Scenario	Actor	System
	1. The traveler selects a destination.	
		2. The system retrieves relevant safety tips and awareness notes from the database.
		3. The system displays the safety tips alongside the destination information.
	4. The traveler reads the displayed safety notes.	
		5. The system allows the traveler to view additional tips
Exceptional Scenario (Alternative flow)	Actor	System
	1. The traveler selects a destination.	
		2. The system fails to retrieve safety tips due to missing data or network error.
		3. The system displays a fallback message: "Safety tips unavailable at the moment."
		4. The traveler may refresh or retry to load the information.

3.2.2.5 Usecase Scenario #5 <Save Important Safety Reminders>

Use Case Name	Save Important Safety Reminders	
Description	The system shall allow the traveler to save important safety reminders.	
Pre-Condition	The traveler is logged into the system and viewing safety information.	
Post-Condition	The system stores the selected reminders successfully for future access.	
Main Scenario	Actor	System
	1. Travelers view safety information on the page.	
		2. The system displays the option "Save Reminder."
	3. Traveler clicks the save button.	
		4. The system saves the reminder to the traveler's account.
	5. Traveler sees confirmation.	
		6. The system shows a message: "Reminder saved successfully."
Exceptional Scenario (Alternative flow)	Actor	System
	1. Traveler tries to save a reminder.	
		2. The system fails to save due to missing data or connection issues.
		3. System displays an error: "Failed to save reminder. Please try again."

3.2.2.6 Usecase Scenario #6 <Display Short Safety Quotes>

Use Case Name	Display Short Safety Quotes	
Description	The system displays a short, awareness-promoting safety quote when a user first opens the website.	
Pre-Condition	The user has a web browser and an internet connection.	
Post-Condition	The user sees the website's homepage, and a safety quote is displayed.	
Main Scenario	Actor	System
	1. The user opens their web browser and navigates to the website's main URL.	
		2. The system loads the main homepage content.
		3. The system retrieves a safety quote from its database.
		4. The system displays the homepage, including the safety quote, to the user.
Exceptional Scenario (Alternative flow)	Actor	System

	1. The user opens their web browser and navigates to the website's main URL.	
		2. The system loads the main homepage content.
		3. System attempts to retrieve a safety quote but fails (e.g., database error, no quotes available).
		4. The system continues to load and display the main homepage without the safety quote.

3.2.2.7 Usecase Scenario #7 <View Interactive Risk Map>

Use Case Name	View Interactive Risk Map
Description	The system shows an interactive map where geopolitical risk areas are differentiated by color shading.
Pre-Condition	The user is logged in (or on a public-facing page) and has navigated to the map view.

Post-Condition	The user can see and interact with the map, which displays color-coded risk levels.	
Main Scenario	Actor	System
	1. The user navigates to the page containing the risk map.	
		2. System retrieves base map data from a map API.
		3. The system fetches the latest geopolitical risk data such as risk levels and corresponding locations.
		4. The system renders the map, overlaying the color-shaded risk areas.
	5. User interacts with the map (e.g., zooms, pans, or clicks on a shaded area)	
		6. The system

		responds to the interaction (e.g., zooms in, displays a pop-up with details about the selected area's risk level).
Exceptional Scenario (Alternative flow)	Actor	System
	Case A: Map data fails to load	
	1. User navigates to the page containing the risk map.	
		2. System fails to retrieve base map data from the map API.
		3. System displays an error message, "Map could not be loaded. Please check your internet connection".
	Case B: Risk data fails to load	
	1. User navigates to the page containing the risk map.	
		2. System successfully loads the base map but fails to retrieve the risk data.

		3. System displays the map <i>without</i> the color-shading and shows an error message, "Risk data is currently unavailable".
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3.2.2.8 Usecase Scenario #8 <Register New Account>

Use Case Name	Register New Account	
Description	The system allows a new user to register for a separate account as either a "Traveler" or a "Travel Agency Partner."	
Pre-Condition	The user is not logged in and is on the system's registration page.	
Post-Condition	The system creates a new user account with the specified role, logs the user in, and directs them to their dashboard.	
Main Scenario	Actor	System
	1. User navigates to the "Register" page.	
	2. User selects their desired account role ("Traveler" or "Travel Agency Partner").	
	3. User fills in all required registration fields, such as name, email,	

	password, company info if a Travel Agency Partner.	
	4. User submits the registration form.	
		5. System validates the submitted data (e.g., checks for unique email, password strength).
		6. System creates a new user record in the database, assigning the selected role.
		7. System authenticates the user and creates a new session.
		8. System displays a "Registration Successful" message and redirects the user to the appropriate dashboard (Traveler or Travel Agency Partner)
Exceptional Scenario (Alternative flow)	Actor	System
	1. User navigates to the "Register" page.	
	2. User selects their desired account role ("Traveler" or "Travel Agency Partner").	

	3. User fills in all required registration fields, such as name, email, password, company info if a Travel Agency Partner.	
	4. User submits the registration form.	
		5. System validation fails (e.g., email is already in use, password is too weak).
		6. System re-displays the registration form, highlighting the fields with errors and providing explanatory messages.
	7. User corrects the data and re-submits (returns to <i>Normal Flow of Events</i> , step 4).	

3.2.2.9 Usecase Scenario #9 <Manage User Accounts>

Use Case Name	Manage User Accounts
Description	This use case allows the system administrator to create, update, and delete user accounts to control access, manage credentials, and ensure accurate user information.

Pre-Condition	The administrator should be logged into the system with administrative privileges and valid credentials.	
Post-Condition	The system creates a new user account with the specified role, logs the user in, and directs them to their dashboard.	
Main Scenario	Actor	System
	1. Administrator selects the "Manage User Accounts" option.	
		2. System displays the list of existing user accounts.
	3. Administrator chooses to create, edit, or delete a user account.	
		4. System opens the respective interface (form for create/edit or confirmation for delete).
Exceptional Scenario (Alternative flow)	Actor	System
	1. Administrator enters invalid or incomplete details.	
		2. System displays an error message prompting for corrections.

	3. Administrator retries with valid data.	
		4. System processes the operation successfully.

3.2.2.10 Usecase Scenario #10 <Search Bar for Searching Destinations>

Use Case Name	Search Bar for Searching Destinations	
Description	This use case enables users to search for their desired destinations using a search bar that provides relevant and quick results.	
Pre-Condition	The user must be logged in or on the homepage where the search bar is available.	
Post-Condition	The system displays destination results that match the user's search query.	
Main Scenario	Actor	System
	1. User enters the desired destination in the search bar.	
		2. System processes the input query.
	3. User clicks the search button or presses Enter.	
		4. System retrieves matching destination data from the database.
Exceptional Scenario (Alternative flow)	Actor	System

	2. User navigates to the "Register" page.	
	2. User selects their desired account role ("Traveler" or "Travel Agency Partner").	
	3. User modifies the search query.	
		4. System reprocesses and shows updated results if available.

3.2.2.11 Usecase Scenario #11 <Verifying Information>

Use Case Name	Verifying Information	
Description	The system shall enable an Affiliated Journalist to submit information for verification to a Risk Expert. The system shall notify the Risk Expert of the pending verification request. The system shall allow the Risk Expert to confirm or reject the geopolitical data accuracy.	
Pre-Condition	1) The Journalist is logged into the system 2) The Journalist created a data and marks it as requiring verification 3) The system has at least one available Risk Expert on standby	
Post-Condition	The Journalist's notified of the outcome in which is if the data is already marked as "Verified" or "Need more revision"	
Main Scenario	Actor	System

	1. “Journalist” : Submits the data for verification	
		2. Sends the data to the risk analysts
	3. “Risk Expert” : Checks the data that has been sent by the Journalist sent by the Journalist.	
		4. Displays the data from the journalist to the risk expert to verify
		5. Updates the verified data into the database
		6. Notifies the Journalist that the data has been verified and is ready for final publication.
Exceptional Scenario (Alternative flow)	Actor	System
	1. “Risk Expert” : Decides that the data is inaccurate or requires more information. And sends it to the Journalist	
		2. Displays a text box requiring a reason on why the data is rejected

		3. Updates the data with the mark “Clarification” and sends it into the Journalist’s account
	4. “Journalist” : Receives the marked data to review and modify before sending it back again to the Risk Expert and returns to step 1 of the normal flow	
		5. The flow repeats

3.2.2.12 Usecase Scenario #12 <Data Filter & Sorting>

Use Case Name	Data Filter & Sorting	
Description	The system should allow users to filter and sort destinations or news articles based on risk levels, such as low, medium, or high.	
Pre-Condition	1. The User is viewing a page that displays a list of news articles or destinations. 2. Each item in the list has a defined risk level (Low, Medium, or High).	
Post-Condition	The displayed list of items is immediately refreshed to show only the results matching the selected filters and ordered according to the selected sort criteria.	
Main Scenario	Actor	System
	1. “User” : Navigates to the Destinations List page.	
		2. Displays the default list of all destinations, usually sorted by date

		or alphabetical order.
	3. “User” : Click the "Risk Filter" dropdown menu.	
		4. Displays a list of available risk levels: Low, Medium, High, and "All."
	5. “User” : Selects "Risk Level: High" and then clicks the "Apply" button (or the filter applies instantly).	
		6. Executes a database query to retrieve only destinations with a 'High' risk classification.
	7. “User” : Clicks the "Sort By" dropdown menu.	
		8. Displays sorting options such as "Date: Newest First" and "Risk Level: High to Low."
	9. “User” : Selects "Risk Level:" High to Low."	
		10. Sorts the filtered list (which currently only contains 'High' risk items) by an auxiliary criterion (e.g., Title A-Z) since all risk values are the same.
	11. “User” : Changes the filter to "Risk Level:	

	Medium."	
		12. Executes a new query and displays the list of 'Medium' risk destinations, still sorted by the chosen secondary criteria.
Exceptional Scenario (Alternative flow)	Actor	System
	1. "User": Selects a combination of filters (e.g., "Risk Level: Low" and "Region: Middle East").	
		2. Executes the database query based on both filters. If the query returns zero results, displays an informational message: "No items match your selected filter criteria. Please adjust your risk level or region selection."

3.2.2.13 Usecase Scenario #13 <Administrator data management>

Use Case Name	Managing database content	
Description	Administrators have the authority to manage database content, including adding, updating, and deleting data related to users, destinations, and safety information.	
Pre-Condition	- The actor is connected to the internet. - The actor has logged in to the authorised account.	
Post-Condition	- The actor now has access to the database. - The actor can add, update, and delete data.	
Main Scenario	Actor	System
	1. The administrator is logged in to the authorised account	
	2. “The administrator”: access the database.	
		3. The system will Verify if the account has access.
		4. The system will give access to the users if it’s verified.
		5. The system will display the database.
	6. “The administrator”: now has access to the database and can manage the database	
	7. “The administrator”: add, remove, or delete data that has been verified.	

		8. The system will check if the action has been verified.
		9. The system will change the content of the database
	10. “The administrator”: can also remove authority for users.	
		11. The system will check if the action is verified
		12. The system will remove the authority from a user.
Exceptional Scenario (Alternative flow)	Actor	System
	1. “The administrator”: logged in to the authorised account	
	2. “The administrator”: tries to access the database.	
		3. The system will verify if the account has access
		4. The system doesn’t give access to the user.
		5. The system will display “User

		doesn't have access."
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3.2.2.14 Usecase Scenario #14 <Travel route planning assistance>

Use Case Name	Travel route planning assistance	
Description	Travel agency partners will have access to an intelligent route planning feature that suggests optimal travel routes and destinations based on geopolitical conditions and safety data.	
Pre-Condition	- The travel agency must be affiliated with the website. - The actor must have access to the internet - The actor is logged in to the authorised account (FR-15).	
Post-Condition	- The actor will have access to an optimal route provided by the website.	
Main Scenario	Actor	System
	1. The actor is logged in to an authorised account (FR- 15)	
	2. The actor goes to a specialized menu	

	for travel agencies, “Travel route.”	
		3. The system will display the “Travel route” menu.
	4. The actor will search for the location they want to find a route for (FR-10).	
		5. The system will provide a map of the location.
	6. The actor will activate the “Generate Route”	
		7. The system will provide the optimal route for the location based on geopolitical conditions and safety data.
		8. The system will provide information and news regarding the location
	9. The actor can also request other alternatives	
		10. The system will provide other optimal alternatives.
	11. The actor can save the route and information.	

	12. The actor can use the route and information for their agency.	
Exceptional Scenario (Alternative flow)	Actor	System
	1. The actor is logged in to an authorised account (FR-15)	
	2. The actor goes to a specialized menu for travel agencies, “Travel route.”	
		3. The system will display the “Travel route” menu
	4. The actor will search for the location they want to find a route for(FR-10)	
		5. The system will provide a map of the location.
	6. The actor will activate the “Generate Route”	
		7. The system doesn’t have an optimal or

		safe route for the location.
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3.2.2.15 Usecase Scenario #15 <Application Login>

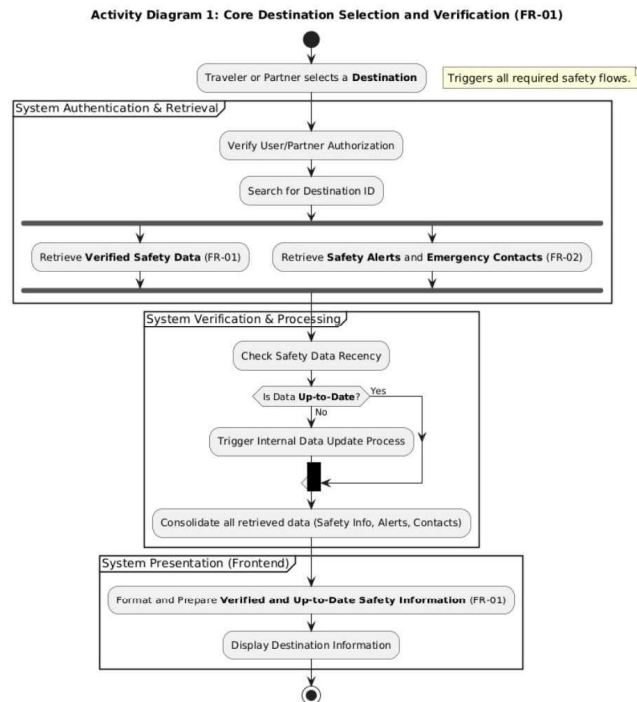
Use Case Name	Application Login	
Description	The system shall provide separate login and registration functionality for the Traveler role. The system shall provide separate login and registration functionality for the Travel Agency Partner role.	
Pre-Condition	- The User has a registered account and valid login credentials. - The User is viewing the Login Page. - The system is connected to the user database.	
Post-Condition	The User is authenticated, logged in, and successfully directed to their role-specific landing page. A session is established.	
Main Scenario	Actor	System
	1. "Users": Enter their email address and password into the respective fields.	
		2. Performs basic input validation (e.g., checks for valid email format,

		non-empty fields).
	3. "User": Clicks the "Login" button.	
		4. Send the credentials to the authentication service.
		5. Verifies the credentials against the user database.
		6. Checks the User's assigned Role (e.g., Traveler, Partner, Admin).
		7. Creates a secure session token and redirects the user.
	8. "Traveler": Is successfully authenticated.	
		9. Displays the Traveler Dashboard (e.g., My Bookings, Saved Destinations).
	10. "Travel Agency Partner": Is successfully authenticated.	
		11. Displays the Partner Portal (e.g., Client Management, Booking Reports).
	12. "Administrator": Is successfully authenticated.	
		13. Displays the Administrator Panel

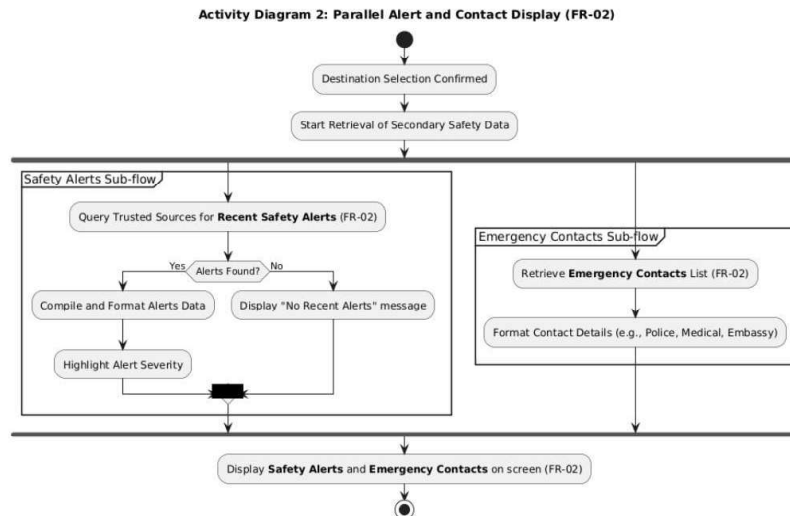
		(e.g., Content Management, User Management).
Exceptional Scenario (Alternative flow)	Actor	System
	1. "User": Enters an incorrect password or an email address not found in the database.	
		2. The authentication service rejects the credentials.
		3. Displays an error message on the login page: "Invalid email or password. Please try again."
	4. "User": Tries to log in again.	
		5. The flow returns to the start of the Normal Flow.

3.2.3 Activity Diagram

3.2.3.1 Activity diagram #1

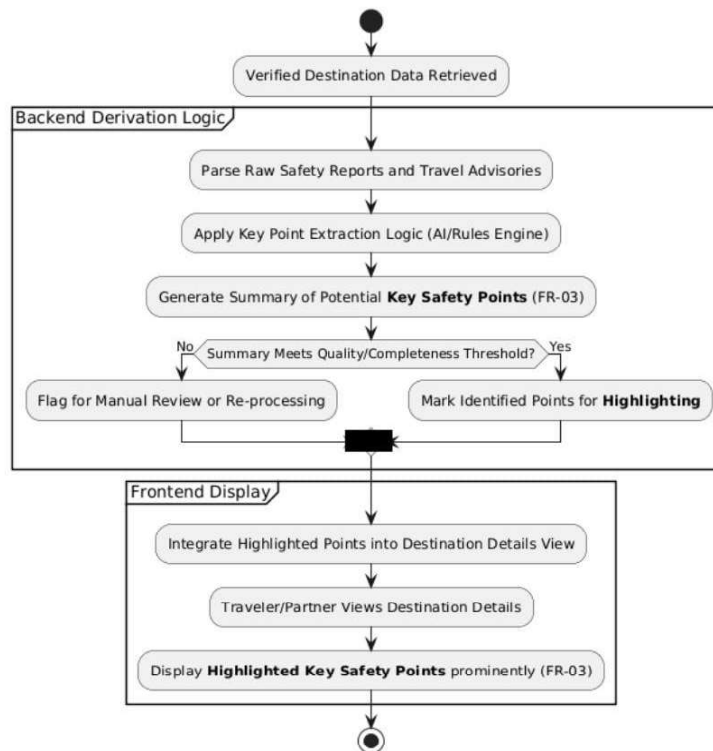


3.2.3.2 Activity Diagram #2

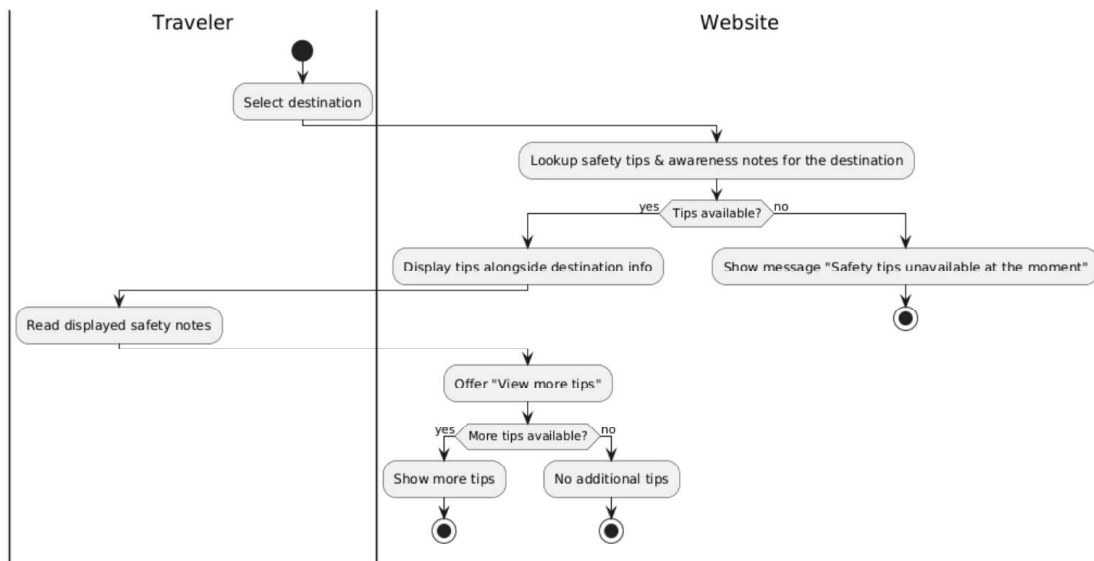


3.2.3.3 Activity Diagram #3

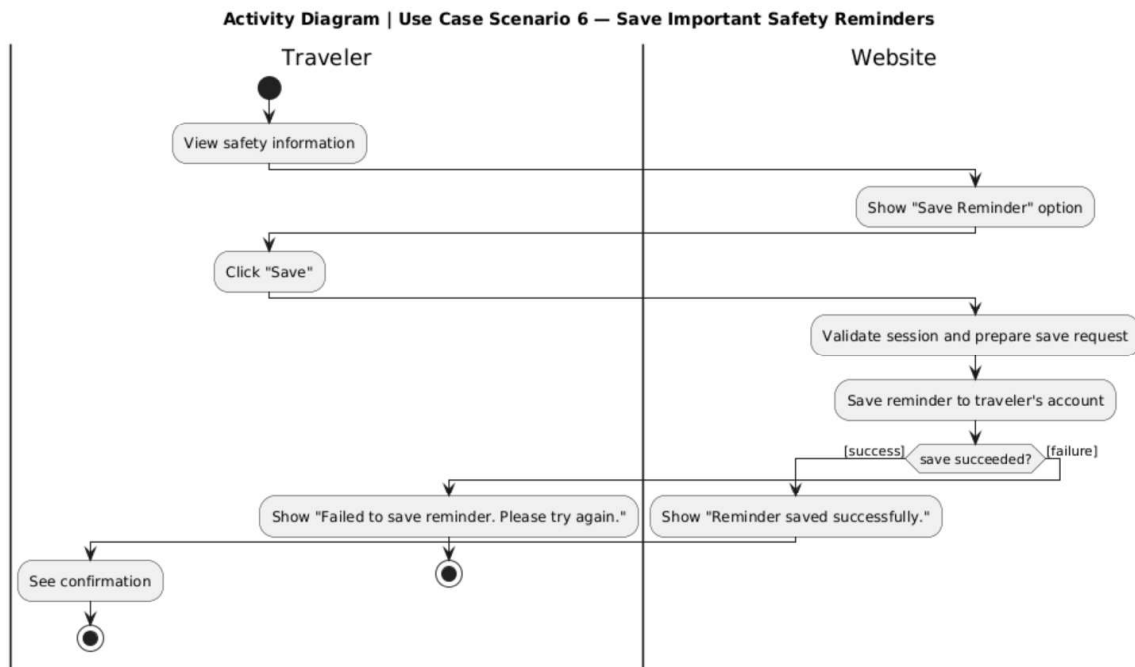
Activity Diagram 3: Key Safety Point Derivation and Highlighting (FR-03)



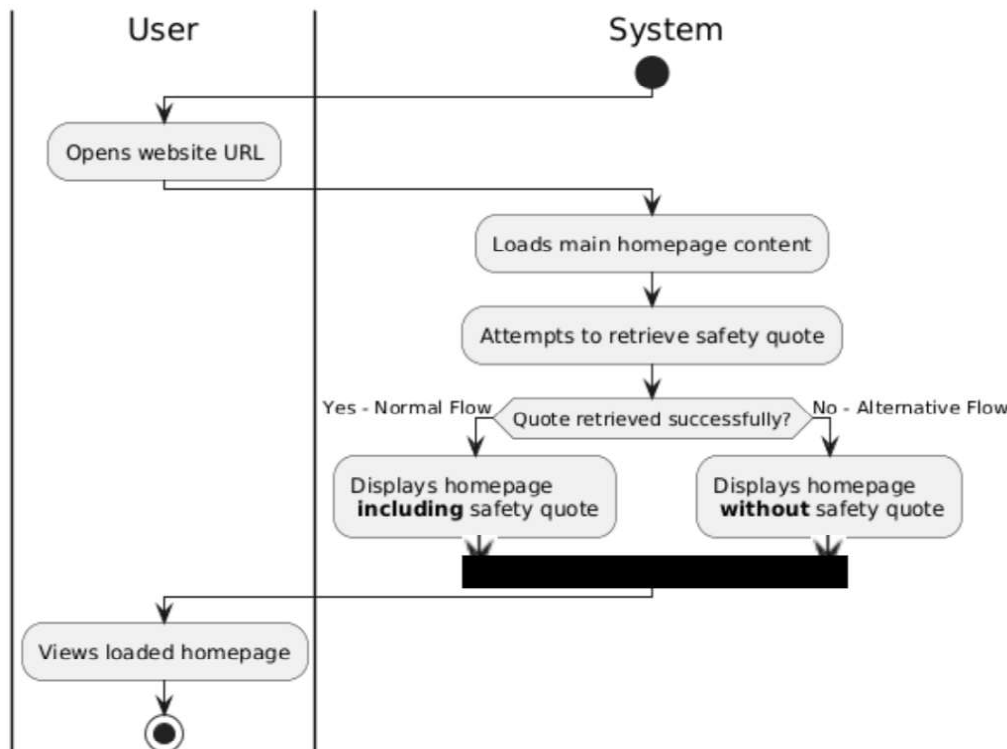
3.2.3.4 Activity Diagram #4



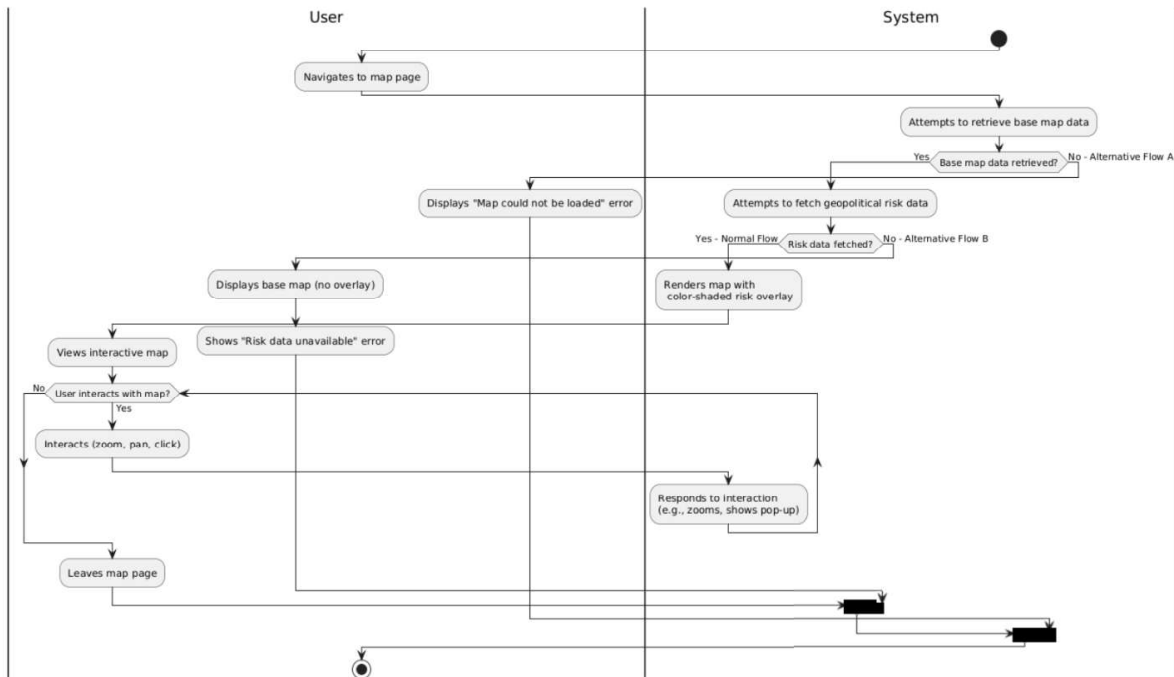
3.2.3.5 Activity Diagram #5



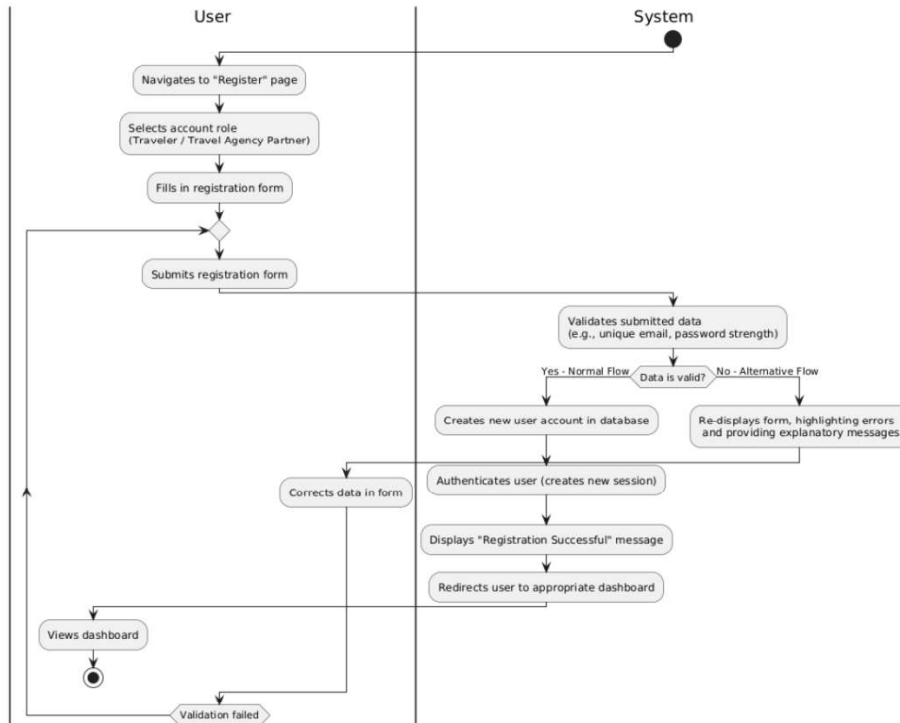
3.2.3.6 Activity Diagram #6



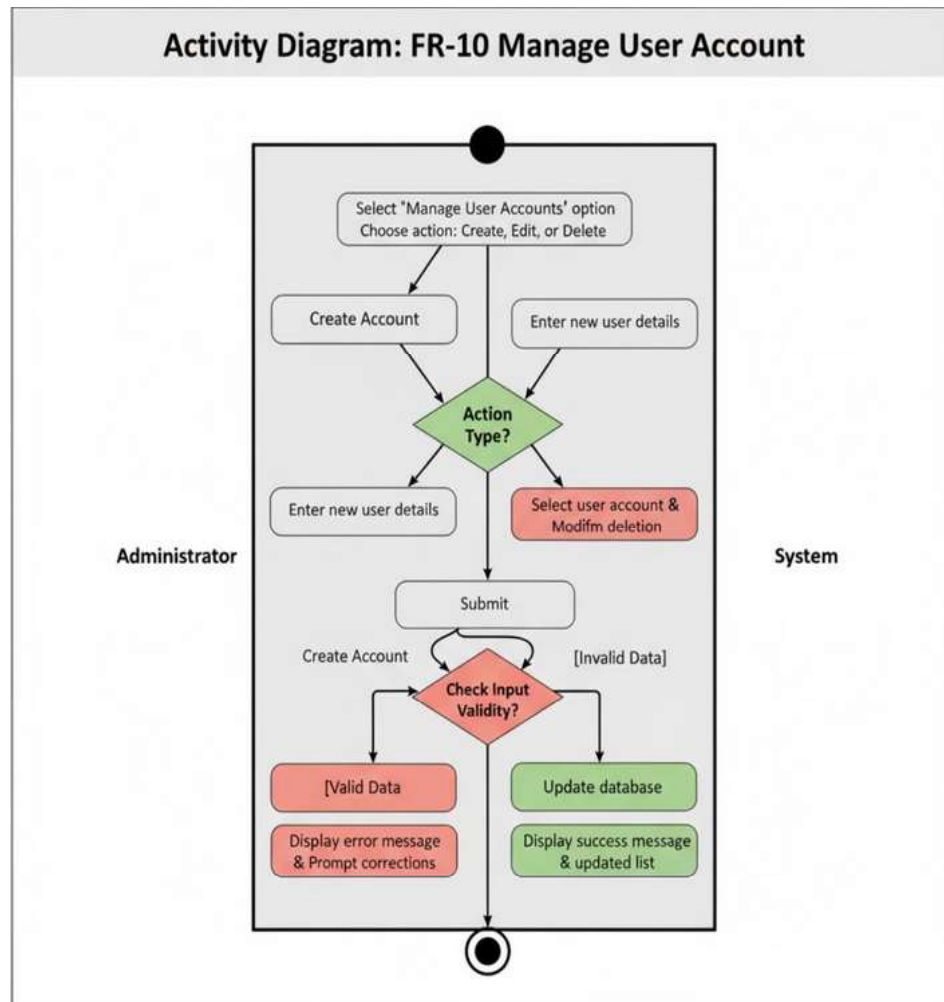
3.2.3.7 Activity Diagram #7



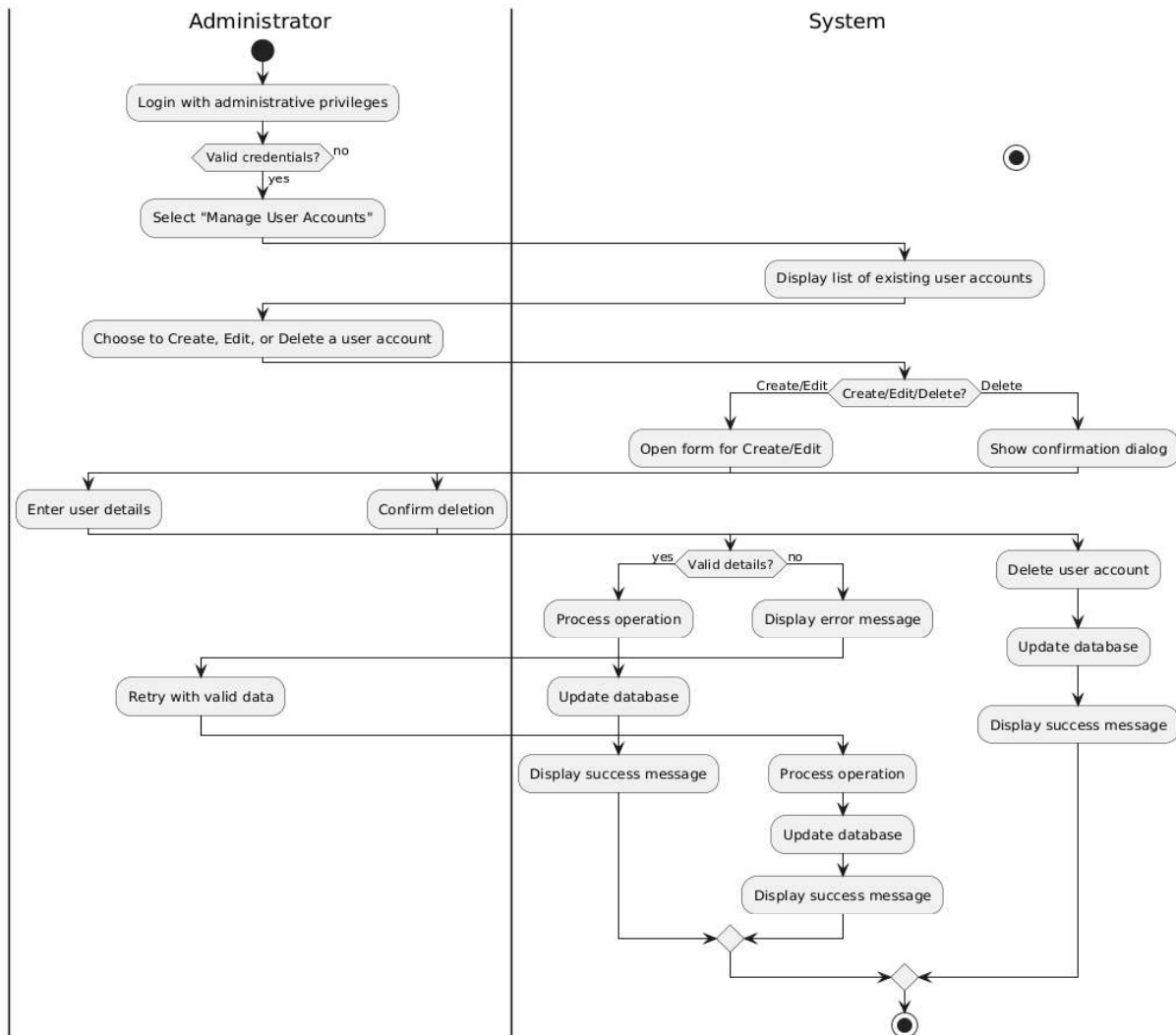
3.2.3.8 Activity Diagram #8



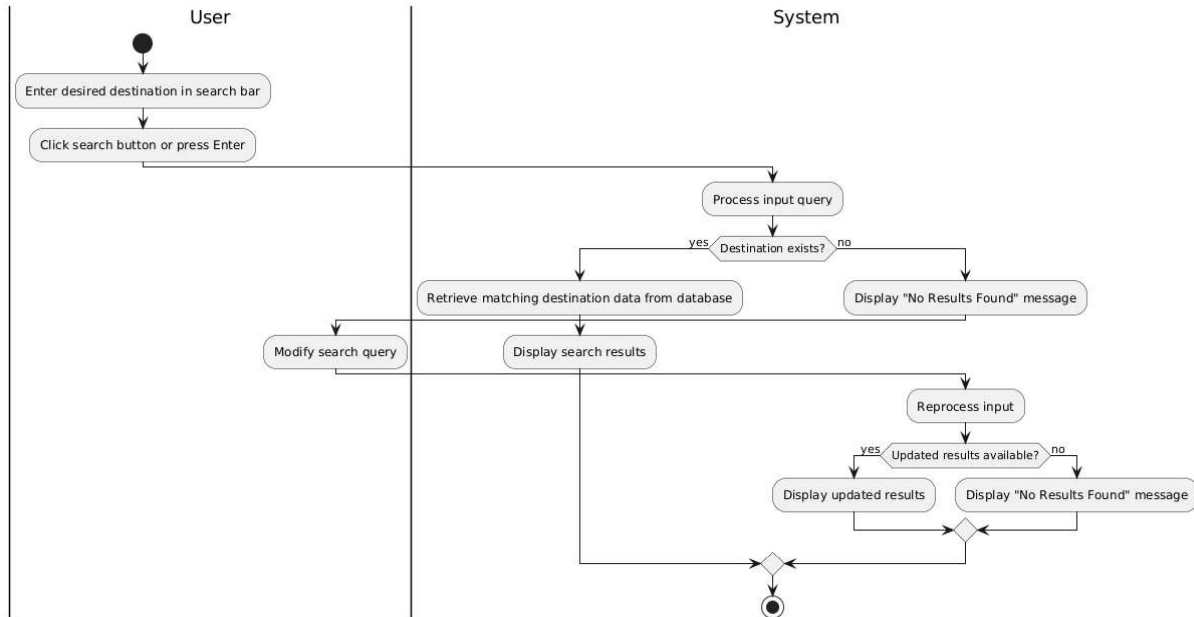
3.2.3.9 Activity Diagram #9



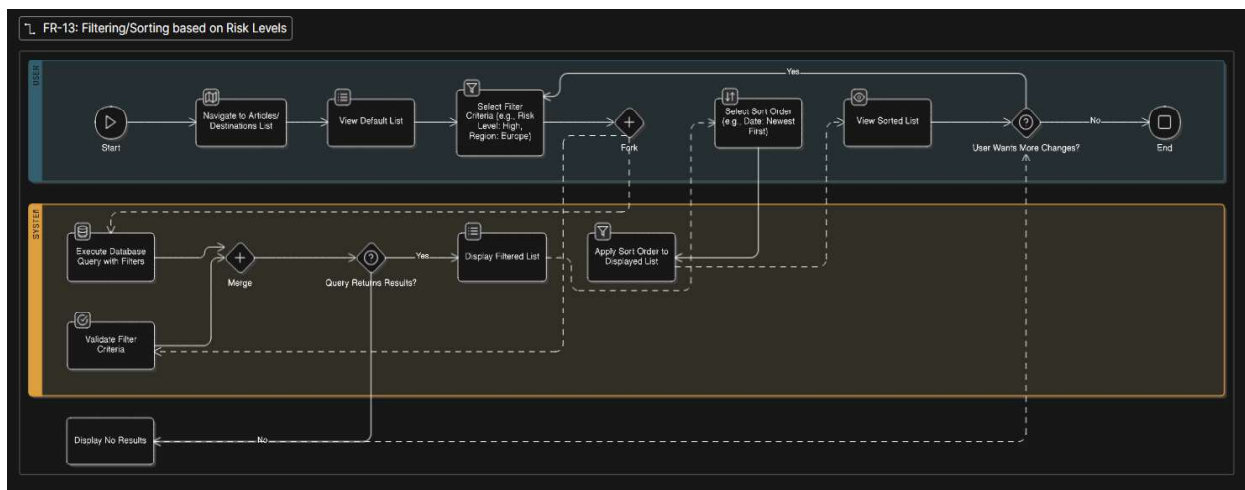
3.2.3.10 Activity Diagram #10



3.2.3.11 Activity Diagram #11

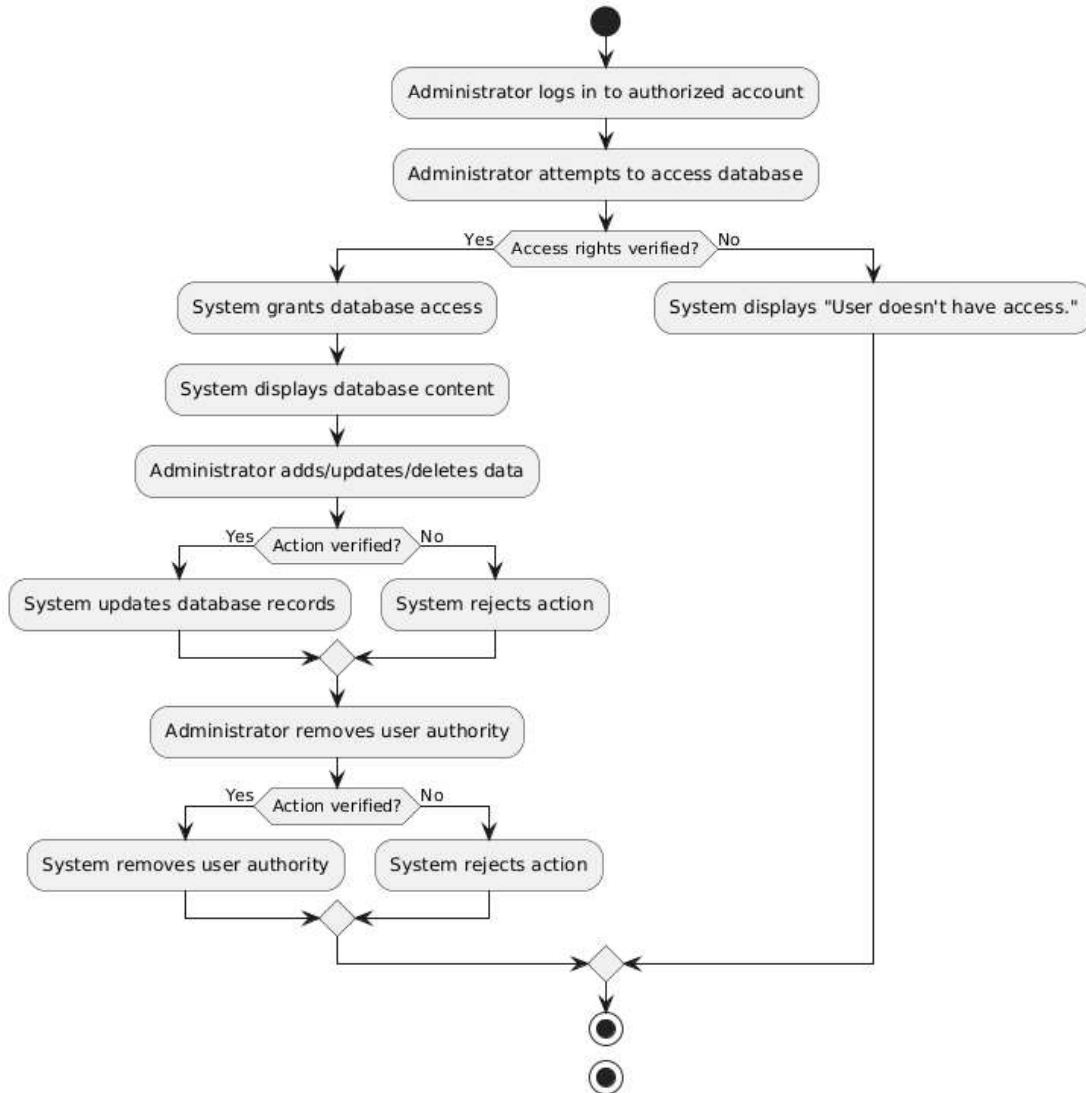


3.2.3.12 Activity Diagram #12

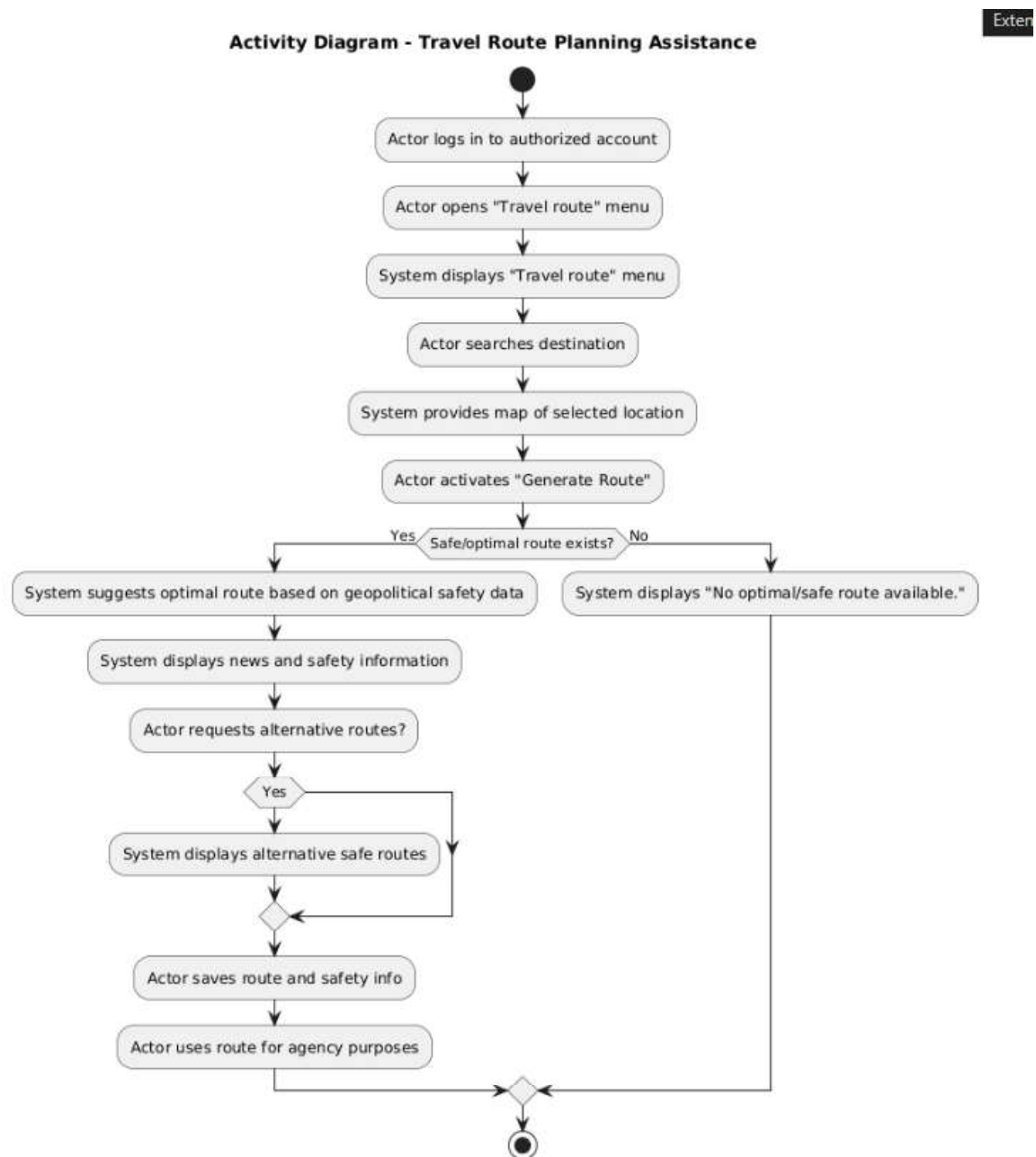


3.2.3.13 Activity Diagram #13

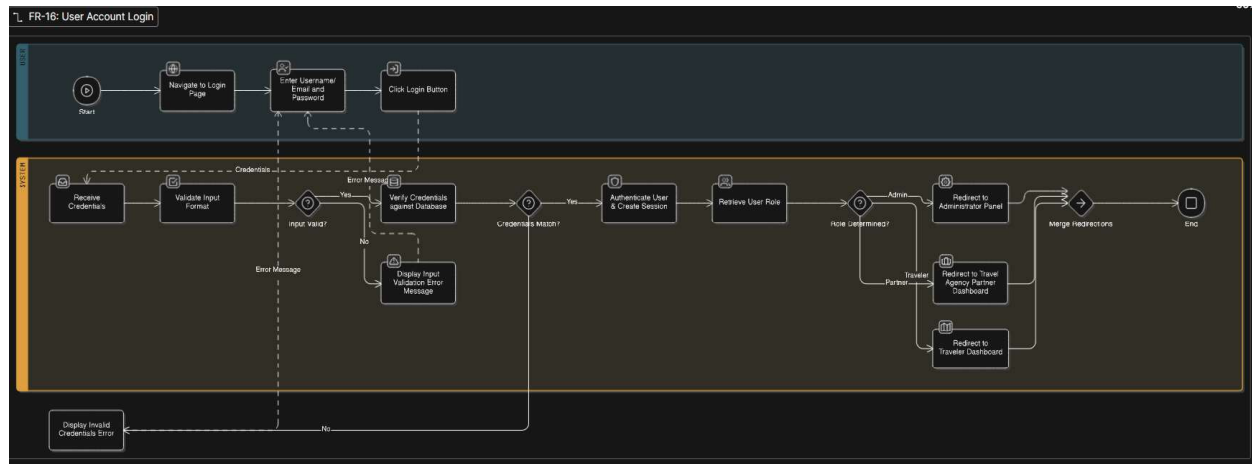
Activity Diagram - Administrator Data Management



3.2.3.14 Activity Diagram #14



3.2.3.15 Activity Diagram #15



3.2.4 Class Diagram

The following figure is the class diagram.

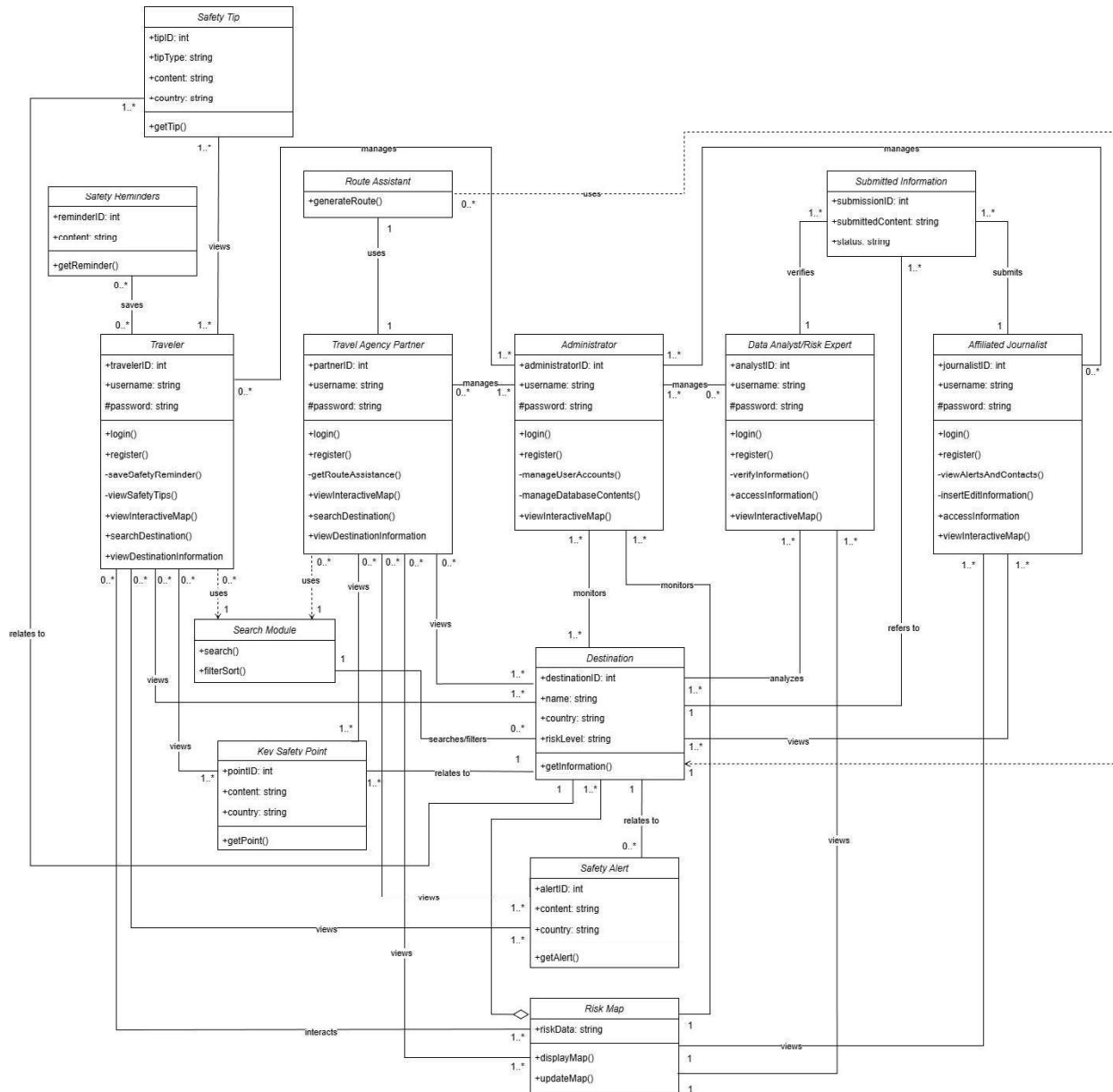


Figure 3.2.4.1 Class Diagram

4. Other Requirements

4.1 User Interface

The User Interface (UI) must be designed to be intuitive and accessible for all user classes. To achieve this, the system must provide a visually appealing UI that enables all travelers to easily navigate and access main features with minimal effort. As emphasized by user feedback, the UI must be accessible, user-friendly, and well-engineered. Given the large amount of information the website will present, its structure must be logical and easy for users to navigate. A key UI component will be an interactive and colorful map design, which will use color shading to differentiate geopolitical risk areas. Furthermore, the UI must be tailored to support the primary activities of its distinct user classes, allowing Travelers to easily view risk maps and alerts while providing Travel Agency Partners with an interface that supports their workflow for assessing regional risks and planning secure routes. The following figure 4.1.1 is a screenshot from the U.S. travel advisories website as a reference example of the travel advisory map concept similar to the interactive risk map that Sentinel will provide.

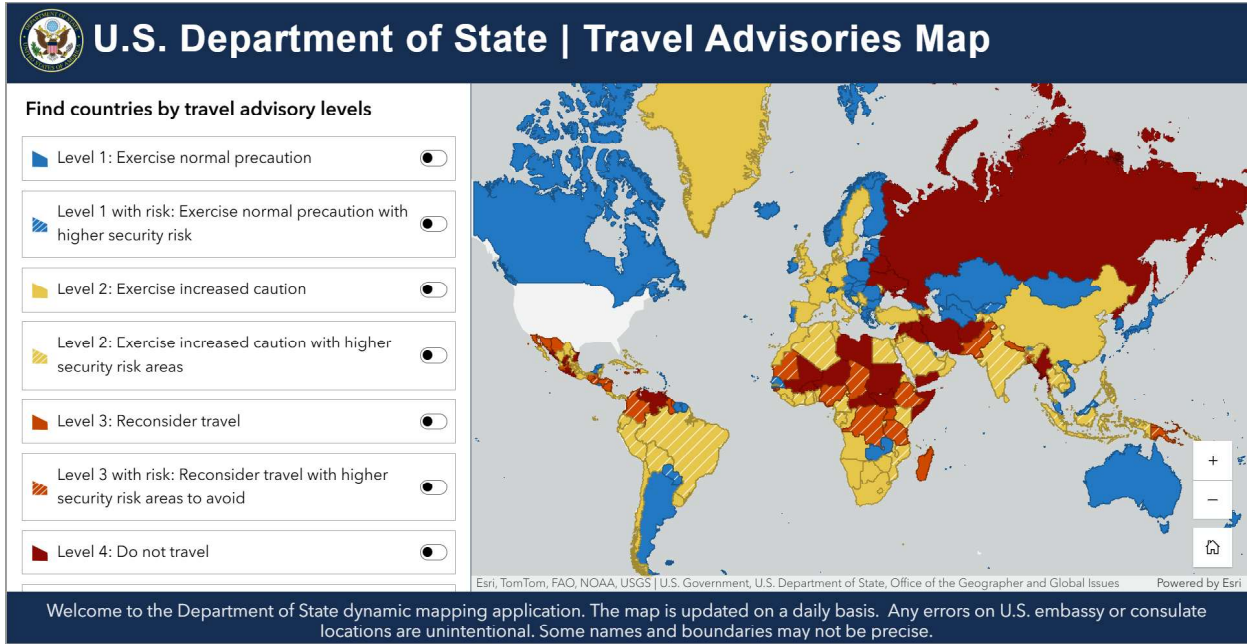


Figure 4.1.1 Reference for interactive risk map.

4.2 Hardware Interfaces

Sentinel is a fully web-based system and does not require any specialized hardware. The platform operates on standard user devices such as laptops, desktops, tablets, or smartphones

with a modern web browser and stable internet connection. No additional hardware components are needed for system functionality.

4.3 Software Interfaces

The Sentinel platform interacts with external software services to deliver real-time and reliable travel information. To support features described in FR-01 (Destination Information), FR-02 (Safety Alerts), and FR-07 (Interactive Risk Map), the system retrieves data from government travel advisory APIs, public health sources, news feeds, and a map API provider. All data exchanges use standard web formats such as JSON over HTTPS to ensure secure and consistent communication. The backend then validates and processes the incoming information so the system remains accurate and stable across all integrated components.

4.4 Communication Interfaces

The Sentinel platform relies on secure and reliable communication to deliver real-time travel information. All interactions between the user's device and the server take place through HTTPS to ensure encrypted data transmission. The system uses API-based requests to obtain external travel advisories, safety alerts, and map data, while the backend returns processed information in standardized formats such as JSON. Consistent internet connectivity is required to maintain timely updates and support features such as FR-02 (Safety Alerts) and FR-07 (Interactive Risk Map).

4.5 Intelligent System Features

The system will incorporate intelligent features to provide advanced decision support and enhance user value. A primary intelligent feature is specified in FR-14, Travel route assistance, a "Delighter" requirement. This feature shall provide Travel Agency Partners with an intelligent route planning capability that suggests optimal travel routes based on real-time geopolitical conditions and safety data. Furthermore, user elicitation identified a strong desire for additional intelligent capabilities. Interview feedback, for example, described the ideal tool as an intelligent assistant. This assistant would provide a structured platform allowing users to filter complex information and easily compare different options, thereby helping them make more confident and informed travel plans.

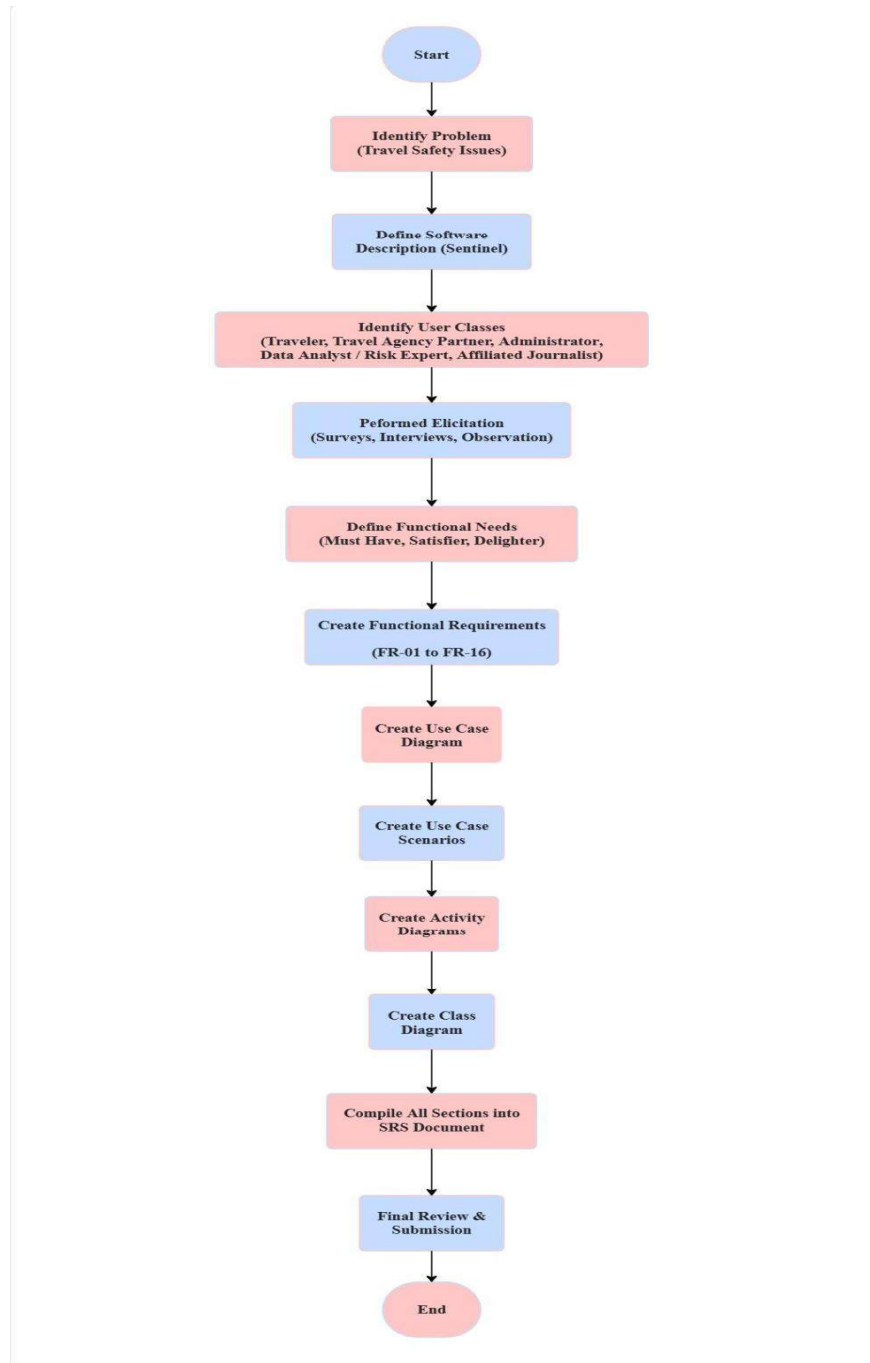
Appendix A: Glossary of Terms

This glossary defines key terms used throughout the Sentinel SRS to ensure clarity and consistent understanding of system concepts.

TERM	DEFINITION
Sentinel	“Sentinel” is an intelligent travel-safety advisory platform that centralizes verified geopolitical data, real-time alerts, and official travel advisories to help users make informed and safer travel decisions for any chosen destination.
Safety Alert	A real-time notification displayed on the platform when urgent issues arise, such as disasters, conflict escalation, or emergencies at a selected destination.
Geopolitical Risk	Any threat originating from political instability, conflict, terrorism, civil unrest, or other factors that may endanger travelers.
Interactive Risk Map	A visual map in Sentinel that uses color-coded shading to display the geopolitical risk level of countries or regions. (As defined in FR-07).
Verification Workflow	The process where journalists submit data, risk experts verify or reject it, and verified information is published on the platform (use case 11).
Reminder	A saved safety tip or key point that a traveler stores for future use (FR-05).
Destination Information	The combined safety data, alerts, risk levels, cultural notes, and emergency contacts displayed when a user selects a destination (FR-01, FR-02, FR-03, FR-04).
Search Bar	A feature allowing users to search for destination information quickly (FR-10).
Optimal Route	A suggested travel route based on real-time safety and geopolitical conditions, available to Travel Agency Partners (FR-14).

Appendix B: Analysis Models

The flowchart below summarizes the key steps taken in developing the Sentinel system, from identifying the problem to completing the final SRS.



Attachment: List of Group Members and Contribution

The following table lists the members of Group Anti-Heroes and outlines their respective contributions to the development of this SRS document.

#	Full Name	Student ID	Contribution
1.	Farrell [REDACTED] [REDACTED]	[REDACTED]	- Introduction (1.3 - 1.4) - General Software Description (2.1 - 2.3) - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-11, FR-12, FR-15
2.	Putu Argya [REDACTED]	[REDACTED]	- General Software Description (2.4 - 2.6) - Analysis Modelling (3.2.1 Use Case Diagram) - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-13, FR-14
3.	Axella [REDACTED] [REDACTED]	[REDACTED]	- Revision History - List of Changed Pages - Table of Contents - Analysis Modelling (3.2.3 Class Diagram) - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-06, FR-07, FR-08
4.	Anita [REDACTED] [REDACTED]	[REDACTED]	- Other Requirements (4.1 - 4.5) - Appendix A & Appendix B - Attachment - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-04, FR-05
5.	Chahat	[REDACTED]	- Introduction (1.1 - 1.2) - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-09, FR-10
6.	Naisha [REDACTED]	[REDACTED]	- Detailed Software Description (3.1) - Individual Contribution: 3.2.2 & 3.2.3 - FR Responsibility: FR-01, FR-02, FR-03